

Subcritical Crack Propagation in Solder Resist Materials During Fatigue Bulge Testing in Experiment and Simulation

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Abstract—Flip-Chip Ball Grid Array (FCBGA) packages, which offer higher integration density and electrical performance compared to classical wire bonding, are increasingly used in automotive electronics and advanced computing applications. As the demand for compact and high-performance electronics accelerates, FCBGA designs have evolved with finer bump pitches, high-density interconnects, and multi-layered substrates. However, this drive for miniaturization and increased functionality has increased thermo-mechanical reliability challenges, especially under harsh conditions. The AEC-Q100 Grade 0 standard for automotive industry now requires FCBGA packages to withstand higher thermo-mechanical stress over wider temperatures ranges (-55°C to 175°C) during Temperature Cycling (TC), and 175°C during High Temperature Storage (HTS). Under these conditions, a critical failure mechanism is the formation and propagation of microcracks in packaging materials, particularly at the solder resist (SR) adjacent to solder bumps, which could lead to solder extrusion and bridging between copper pillars, causing electrical shorts and device failure. To address these challenges, this study integrates experimental bulge testing, with finite element methods (FEM) to investigate crack growth in SR materials. By analyzing stress distribution and fatigue properties, the research aims to develop accurate fatigue lifetime models and establish robust design guidelines to enhance the reliability of FCBGA packages.

Keywords— Flip-Chip Ball Grid Array, microcontrollers, crack growth, lifetime modeling, Finite Element Modeling, tensile testing, digital image correlation.

I. INTRODUCTION

The automotive industry is undergoing a significant transformation driven by innovations such as autonomous driving, advanced driver-assistance systems (ADAS), and connected vehicle technologies. These advancements have led to increased integration of electronics in vehicles and a growing demand for miniaturized high-performance ICs with higher I/O densities and thinner form factors to enable more intelligent, sophisticated automotive systems. Flip-Chip Ball Grid Array (FCBGA) packaging is a key enabler, providing

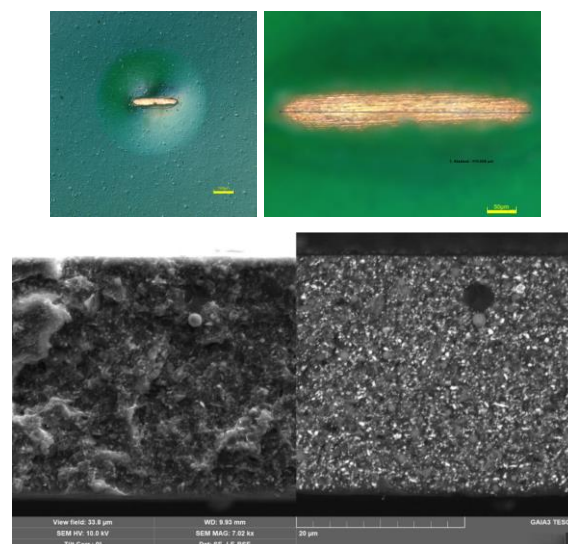


Fig. 1. Solder resist sample with exemplary pre-crack of 400 μm length (top left) prepared by laser, with enlarged view on the right SEM of the SR cross section (bottom): left: fracture surface, right: before fracture, nearly homogeneous matrix with uniformly dispersed filler particles [22].



Fig. 2. CAD model showing detailed construction of the bulge tester sample holder. The assembly consists of viewing glass windows, top and bottom pressure chambers, air inlets/outlets, O-rings for airtight seal and placeholders for thermocouple, cartridge heater and pressure sensor. It facilitates controlled testing by regulating airflow, temperature, and pressure.

high-density interconnects and higher electrical performance suited to the evolving automotive requirements. However, the shift toward more complex electronics, especially in harsh underhood environments, requires packages that can withstand extreme thermal and mechanical stresses. Thermo-mechanical reliability has therefore become highly challenging, particularly to meet automotive standards, such as AEC-Q100 Grade 0, which requires FCBGA packages to maintain functional at storage temperatures upto 175°C, exceeding the conventional threshold of 150°C. In substrates with smaller bump pitches, thermal stresses in the solder resist (SR) — critical for insulation and structural integrity — tend to increase under extended thermo-mechanical storage and thermal cycling and have recently been found [1],[2] to precede chip interconnect failure.

To ensure the durability and reliability of IC packages in automotive environments, new material-specific fatigue life and failure models are essential. Advanced material characterization and finite element modeling (FEM) are used to characterize the material fatigue properties and to simulate crack initiation and propagation, and assess thermo-mechanical responses under reliability tests. The methodology presented in this paper combines experimental bulge testing [3]–[10] precise crack growth measurement, and advanced FEM to understand the crack initiation and propagation criteria in SR materials for contemporary FCBGA microcontrollers.

A key challenge in such material characterization is that conventional fatigue tests are unsuitable to handling thin, brittle specimens due to thickness and stiffness limitations. The use of compact tension (CT) specimens is not feasible. To address this, a specimen-centered approach is adopted. Although bulge testing is a well-established method for thin films to yield elastoplastic properties, it has rarely been used to induce fatigue damage and extract subcritical fracture-mechanics data. In this work, a custom bulge tester is used to characterize subcritical thermo-mechanical fatigue in thin SR films.

While the present study focuses on SR fracture at room temperature, extending testing at elevated temperatures up to 200°C, remains a key objective. Characterizing fracture behavior across loading rates and temperatures, accounting for thermal aging, will enable extraction of fracture parameters to be employed in package-level, multi-material FEM models, to support reliability predictions under thermo-mechanical stress.

II. EXPERIMENTAL SETUP

A. Sample preparation

Free-standing square SR specimen of size 30 mm x 30 mm with thickness 30 μ m were fabricated from laminated SR sheets on a copper film over glass core substrate using laser. The SR samples are then peeled off from the copper layer. Pre-crack of few 100 μ m is introduced at the center of the specimen by 1064 nm ytterbium fiber laser scribing (Fig. 1), which provide precise micromachining of notch geometry for fracture studies. The resulting pre-crack lengths are quantified using optical microscopy (TABLE I.).

B. Experimental Setup and Procedure

Figures 2-4 depict the two-chamber bulge tester setup, comprising two cylindrical pressure chambers of 30 mm diameter and two anti-reflective coated borosilicate glass

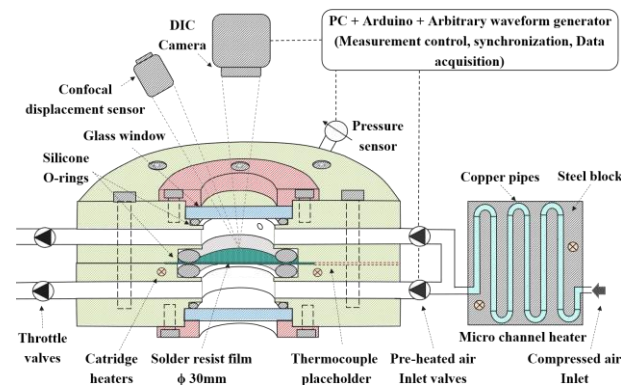


Fig. 3. Schematic cross-section of the Bulge Tester setup. Explanations: see main text.

windows of 30 mm diameter and 3.3 mm thickness mounted at the top and bottom. Square SR specimen of 30 mm length is aligned at the center of the bulge tester chamber across the central sample fixture and clamped along its perimeter using silicone O-rings, enabling application of static or cyclic tensile loading by imposing pressure differential, that bulge the membrane in axial direction of the aperture.

Solenoid valves (Bürkert, 2/2-way, Type 6027), driven by function generator enable both ramped and cyclic pressure control from either sides of the membrane. Pressure (0– 10 bar) is measured using Keller Series M5HB pressure transducers (0.1% accuracy), that provide a standardized 0– 10 V output. The analog signal is digitized and converted into pressure using an Arduino, which also acquires continuous temperature readings from the chamber. The design and fabrication of the bulge tester sample holder was validated using thin membrane pressure-deflection relations (Sec. III.A) and FEM (Sec. III.C). The assembled system has been verified up to 8 bar pressure.

For elevated temperature measurements, a microchannel heater preheats the compressed air while cartridge heaters regulate the sample holder temperature, which is monitored using a thermocouple and controlled via Arduino. An insulated enclosure for the test setup is under development.

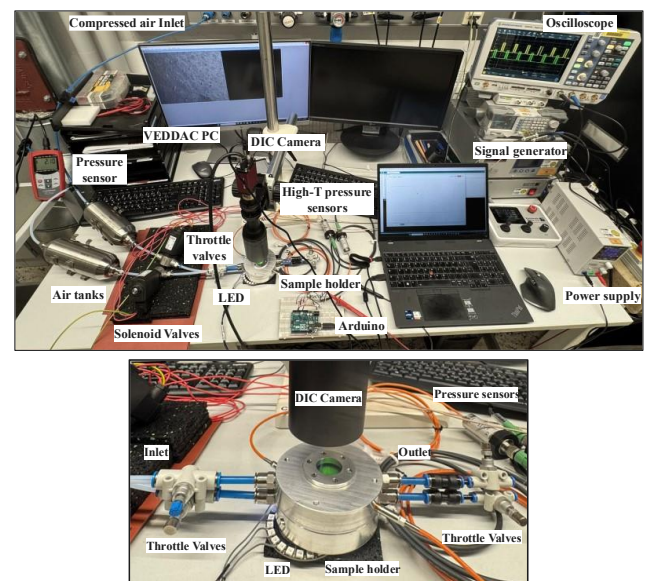


Fig. 4. Overview of current Bulge tester setup (top) and enlarged view of the sample holder pressure chamber in operation (bottom).

To study crack propagation in brittle thin films such as SR, two steps of measurement are employed. First, a monotonic destructive test is performed on the specimen to determine the tensile mode I fracture toughness K_{Ic} and the corresponding critical energy release rate G_{Ic} . As the specimen undergoes out-of-plane (y-direction) displacement under pressure, it develops in-plane strain and the stress at the pre-crack tip increases. Crack extension initiates when the applied stress intensity factor K_I exceeds K_{Ic} (equivalently when $G_I > G_{Ic}$). Accordingly, a pressure ramp is applied to the underside of the membrane to investigate its mechanical behavior under uniaxial loads and determine K_{Ic} (see Section III.B). The membrane is pressurized until failure. K_{Ic} is calculated at the failure pressure using the corresponding crack-tip stress and the measured pre-crack length based on Irwin's solution for a through-thickness crack in an infinite plate (Sec. III.C).

In the second step, the fatigue behavior of the SR membrane is investigated. Sinusoidal, alternating pressure is applied to opposite sides of the membrane by actuating the magnetic valves with a function generator. The resulting displacement field and subcritical crack growth during cyclic loads are captured using a VEDDAC camera system (CWM, Germany) with grayscale digital image correlation (DIC). For each acquired frame, the current crack length a is computed by correlating with the preceding frame using a custom VEDDAC Python script that utilizes the effective y-direction strain to locate the crack tip. This enables continuous tracking of crack length as a function of cycle count N , from which the crack growth rate da/dN is derived. As detailed in Section III, da/dN is then correlated with the applied cyclic stress range $\Delta\sigma$ and the corresponding stress intensity range ΔK_I , obtained via analytical relations and numerical simulation (see Section III.B).

III. SIMULATION AND MODELLING OF CRACK GROWTH

A. Analytical modeling of thin membrane deflections

For thin circular elastic membranes as considered in this paper, and small deflections, the following relation between the chamber pressure p and the deflection h at the membrane center applies [8],[9]:

$$P(h) = 4\sigma_r d \frac{h}{r^2} + \frac{8}{3} \frac{E}{(1-\nu)} d \frac{h^3}{r^4}, \quad (1)$$

where r is the film radius, σ_r is the radial residual stress, d is film thickness, E is Young's modulus and ν is Poisson's ratio of the material. The membrane (radial) stress, σ and strain, ε can also be given as,

$$\sigma = \frac{P(h) r^2}{4hd}, \quad (2)$$

$$\varepsilon = \frac{2h^2}{3r^2} + \varepsilon_0, \quad (3)$$

where ε_0 is radial residual strain. Since the membrane is clamped with negligible initial slack or pre-stress, both residual stress and associated residual tension stress are considered negligible.

B. Analytical modeling of stable crack propagation

The stress intensity factor at the crack tip in the center area of the SR membrane is given [11],[12]

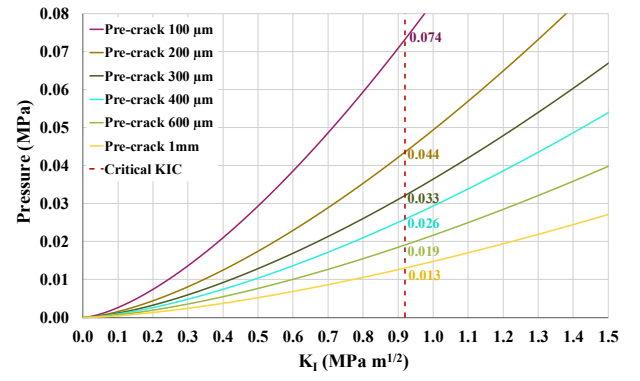


Fig. 5. Crack intensity factor K_I (Mode I) at the crack tip vs. pressure for different pre-crack lengths, evaluated using Eq. (4). The respective pressures (MPa) at which crack initiates are indicated by numbers, assuming fracture toughness of $0.92 \text{ MPa m}^{1/2}$.

$$K_I = \sigma_c \sqrt{\pi a} f\left(\frac{a}{r}\right), \quad (4)$$

where σ_c , a , r denote critical stress, pre-crack length, sample radius and

$$f\left(\alpha = \frac{a}{r}\right) = \frac{1}{\sqrt{1-\alpha}} (1 - 0.5\alpha + 1.6873\alpha^2 - 2.671\alpha^3 + 3.2027\alpha^4 - 1.8935\alpha^5). \quad (5)$$

Fig. 5 shows K_I as a function of applied pressure for various pre-crack length, and indicates the pressure at which crack growth initiates, i.e. when $K_I > K_{Ic}$. The relation between K and energy release rate G depends on sample thickness. For very thin samples (typically $< 0.5 \text{ mm}$) a plane-stress condition can be assumed, giving $G_I = K_I^2/E$.

In the second step, the SR specimen with pre-crack at the center is subjected to cyclic tensile loading in the bulge tester under load-controlled mode at a constant base load of 60–80 % of monotonic failure load and superimposed with a dynamic (sinusoidal) component $\Delta\sigma_c$ to obtain

$$\Delta K_I = \Delta\sigma_c \sqrt{\pi a} f\left(\frac{a}{r}\right). \quad (6)$$

The range $\Delta K_I = K_{I,\max} - K_{I,\min}$ at the crack tip is obtained from Eq. (4) or extracted from FEM for each cycle on the instantaneous crack length a . Depending on the pre-crack length, the cyclic amplitude is selected to ensure subcritical crack growth, i.e. $\Delta K_I > \Delta K_{th}$ while $K_{I,\max} < K_{Ic}$ for each cycle. Under cyclic loading, the crack growth rate (da/dN) correlates with the stress intensity range (ΔK_I) via Paris-Erdogan Law, predicting fatigue life under varying loading conditions [13],

$$\frac{da}{dN} = c \cdot (\Delta K_I)^m. \quad (7)$$

The coefficient c and exponent m depend on the material properties and the temperature. Stable fatigue crack growth is bound below by the threshold ΔK_{th} [14] and above by the fracture toughness K_{Ic} beyond which catastrophic failure occurs. In the sub-critical regime, damage initiates from micro- to nanoscale defects with growth rates on the order of nm per cycle [15]. The acceleration exponent m is obtained from the slope of a linear fit to a log-log plot of measured da/dN vs ΔK_I , where ΔK_I is computed for the test geometry using FEM.

C. Finite Element Modelling

To predict onset of stable or unstable crack propagation under experimental conditions, a dynamic FE model of the bulge fatigue test on a circular SR membrane specimen with defined pre-crack at the membrane center is developed using ABAQUS Standard™.

Explicit dynamic crack growth under cyclic loading is simulation using the eXtended Finite Element Method XFEM (Fig. 6). The SR material was modeled as linear elastic at room temperature [16], [17], with a Young's modulus of 3.8 GPa and a Poisson's ratio of 0.33. The model is used to extract critical pressure required for the pre-crack to propagate. Virtual Crack Closure Technique (VCCT) [18] is employed to simulate crack growth under periodic and ramp-up loading scenarios using critical fracture parameters of $K_{IC} = 0.92 \text{ MPa m}^{1/2}$ and $G_{IC} = 225 \text{ J/m}^2$. Periodic loading is modeled using the direct cyclic step method. Crack propagation is governed by the Benzeggagh–Kenane (BK) criterion [19].

IV. RESULTS AND DISCUSSION

Prior to bulge testing, XFEM simulations are performed to determine the pressure required to initiate crack growth in the SR specimen as a function of pre-crack length. For a pre-crack of 400 μm , the model predicted crack propagation at a relatively low pressure of approximately 0.3 bar under pressure ramp loading conditions, and a vertical membrane displacement of 0.9 mm before abrupt fracture.

Pressure-deflection curves from destructive bulge tests on pre-cracked SR specimen under pressure control are shown in Fig. 7. The measured pre-crack lengths of all destructively tested samples, the corresponding critical pressures at which crack propagates and the calculated fracture toughness according to Eq. (4) are listed in TABLE I. The indicated

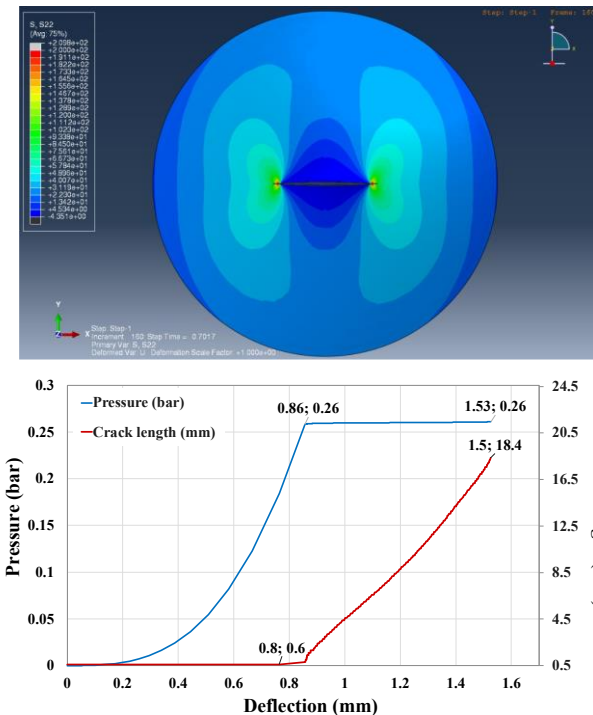


Fig. 6. XFEM model of a SR specimen (pre-crack 400 μm) with Fracture toughness $0.92 \text{ MPa m}^{1/2}$. Simulated deflection (blue) and crack length (gray) under a pressure ramp. Crack propagation occurs at a pressure of ~ 0.3 bar.

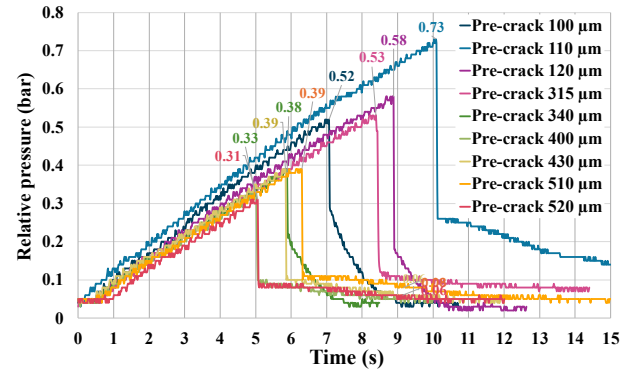


Fig. 7. Destructive pressure ramp tests until membrane rupture. Crack propagation at pressures in the range 0.3-0.6 bar depending on the pre-crack length (100-520 μm).

uncertainty combines both statistical standard deviation and optical measurement uncertainty of the pre-crack length. For the SR, we obtain $K_{IC} = (1.1 \pm 0.2) \text{ MPa m}^{1/2}$ and G_{IC} in a range between 142 and 494 J/m^2 . The calculated fracture toughness for SR is consistent with our initial assumption of $0.92 \text{ MPa m}^{1/2}$ obtained from static FEM of tensile test for fracture characterization on SR [21] (not yet published).

Destructive tests showed crack propagation at ~ 0.4 bar pressure for specimen with 400 μm pre-crack length. This is consistent with ~ 0.3 bar critical pressure inferred from Fig. 6. The corresponding pressure-deflection curves from the destructive tests exhibit smooth pressure drops at failure, however, some exhibit a sharp drop in pressure after crack initiation which might be leading to overshoot in pressure. This behavior is commonly linked to blunt crack tips resulting in overestimated fracture toughness. After further optimizing the pre-crack preparation to get sharper crack tips via FIB milling, and improving the precision of the pressure ramp control, we expect to suppress overshoot and obtain more accurate measurements, characterized by smooth pressure decay at SR membrane rupture and, thus, realistic fracture toughness.

A load-controlled fatigue test was conducted on a standalone SR sample at room temperature with initial pre-crack length of $a_0 = 400 \mu\text{m}$ at the center. The dynamic pressure amplitude, ΔP was set to 0.16 bar, which corresponds to 62 % of the equivalent fracture load, according to Eq. (4). Fig. 8 depicts excerpts out of a VEDDAC DIC sequence showing Region II (stable) crack propagation under cyclic

TABLE I. BULGE TESTS ON SR SPECIMENS: PRE-CRACK LENGTHS, CRITICAL PRESSURE, FRACTURE TOUGHNESS AND CRITICAL ENERGY RELEASE RATE

No.	Pre-crack length (μm)	Critical pressure (bar)	K_{IC} ($\text{MPa m}^{1/2}$)	G_{IC} (J/m^2)
1	100 ± 20	0.52	0.732	142.18
2	110 ± 20	0.73	0.962	245.85
3	120 ± 20	0.58	0.862	197.36
4	315 ± 20	0.53	1.315	459.39
5	340 ± 20	0.39	1.114	329.41
6	400 ± 20	0.33	1.081	310.16
7	430 ± 20	0.39	1.253	416.61
8	510 ± 20	0.39	1.364	494.12
9	520 ± 20	0.31	1.182	370.96

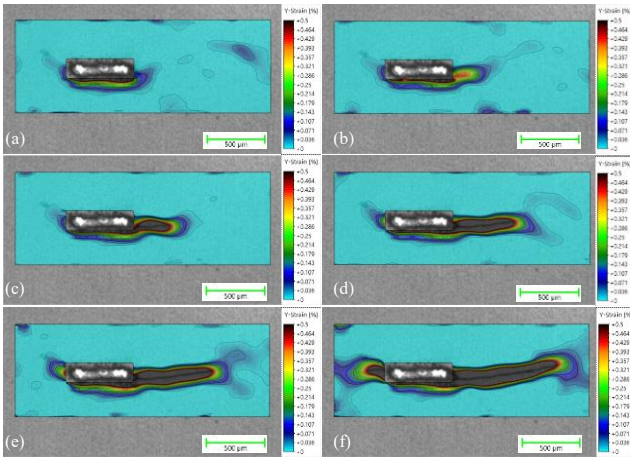


Fig. 8. VEDDAC sequence of a cyclic loading test on SR specimen with pre-crack length of 400 μm : dynamic load $\Delta P = 0.16$ bar. Stable crack growth was recorded after $N = 640$ cycles: (a) Pre-crack opening, 640 cycles, averaged $da/dN = 0.6$ $\mu\text{m}/\text{cycle}$, (b) 422 μm , $N = 730$, 0.6 $\mu\text{m}/\text{cycle}$, (c) 523 μm , $N = 992$, 0.6 $\mu\text{m}/\text{cycle}$, (d) 647 μm , $N = 1100$, 0.6 $\mu\text{m}/\text{cycle}$, (e) 762 μm , $N = 1168$, 0.65 $\mu\text{m}/\text{cycle}$, (f) 1010 μm , $N = 1200$, 0.8 $\mu\text{m}/\text{cycle}$. Remaining images at crack lengths > 1010 μm (unstable crack growth) are not depicted.

loading. The pre-crack started to noticeably propagate after $N = 640$ cycles. Initially, from analytical solution (from Eq. (4)) at corresponding ΔP , $K_{I,\text{max}} = 0.92$ $\text{MPa m}^{1/2}$, which is within the determined fracture toughness range. Stable crack propagation persisted to $N \sim 1200$ cycles, reaching a crack length of ~ 1010 μm . Beyond this point, ΔK_I approaches critical toughness K_{Ic} , and the crack transitioned to the unstable Region III.

For each frame of VEDDAC DIC sequence, the vertical (y) strain field is computed and masked by a chosen strain threshold to isolate the high strain regions associated with the crack. The crack length a and the growth rate da/dN is extracted from the resultant y-strain region using Python. Because the result depends on the chosen y-strain threshold and the DIC grid resolution, the extracted crack lengths have an uncertainty of approximately $\pm 5\%$. Furthermore, we extracted the corresponding ΔK_I values at the crack tip out of the analytical model (Eq. (4)) at the same crack lengths and at nominal load ΔP . Plotting da/dN vs. ΔK_I , normalized to the lower boundary 1.1 $\text{MPa m}^{1/2}$ of fracture toughness in a double-logarithmic graph (see Fig. 10), we determined via a

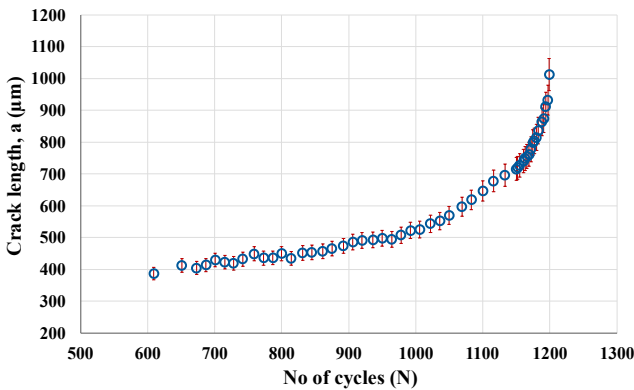


Fig. 9. Crack length extracted from VEDDAC DIC sequence images for each cycles. The pre-crack of length ~ 400 μm started to propagate after $N = 640$ cycles. Stable crack growth persisted to $N \sim 1200$ cycles, reaching a crack length of ~ 1010 μm . Error bars denote $\pm 5\%$ relative uncertainty on the crack-length measurements at each cycle.

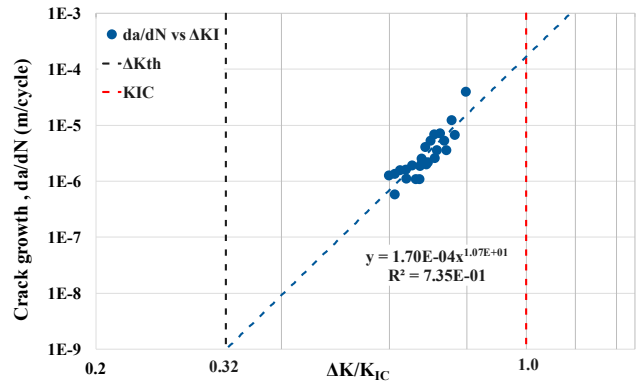


Fig. 10. Paris-Erdogan graph for SR : crack growth rate da/dN vs ΔK_I normalized 1.1 $\text{MPa m}^{1/2}$. The dimensionless Paris-Erdogan exponent (slope of fitted power law trendline) is determined to be $c = 10.7$. No rmalized fracture toughness K_{Ic} and fatigue threshold $\Delta K_{th} = 0.35$ $\text{MPa m}^{1/2}$ (by linear regression to 10^{-9} m/cycles) are indicated by vertical lines.

power-law trendline the dimensionless Paris-Erdogan exponent $m = 10.7$ and the coefficient $c = 170$ $\mu\text{m}/\text{cycle}$ according to Eq. (7). The fatigue threshold $\Delta K_{th} = 0.35$ $\text{MPa m}^{1/2}$ is obtained by linear regression of the trendline to $da/dN = 10^{-9}$ m/cycle, i.e. the order of atomic distances. Close to the fracture toughness K_{Ic} , the crack transitioned to the unstable Region III region and propagated too rapidly to be captured by the available camera frame rate.

V. SUMMARY AND OUTLOOK

Conventional fatigue methods (e.g., compact-tension or dog-bone tests) are difficult to apply to thin, brittle films because of specimen handling, gripping, and thickness constraints. To address this, a specimen-centered approach is adopted. In this paper we have developed and validated bulge tester setup for characterizing fatigue damage properties in SR membranes. We have presented a first assessment of destructive crack growth in SR membranes at room temperature, combined with FEM. A static FEM model provides K_I and G_I values consistent with analytical relations, and an XFEM framework with a BK criterion reproduced crack initiation and growth observed experimentally. From destructive bulge tests we extracted fracture toughness values for SR that are consistent with independent tensile-based estimates, reinforcing the suitability of the bulge testing for thin film characterization. We monitored subcritical fatigue crack growth in-situ under biaxial stress conditions and derived Paris-Erdogan parameters from bulge fatigue measurements.

In future work we will (i) perform bulge measurements across various load rates to quantify rate-dependent behavior, (ii) extend testing to temperatures up to 175°C , above the SR glass-transition temperature using cartridge heaters and microchannel heater, to heat the sample holder and air flowing into it respectively, to capture thermal and viscoelastic effects, and (iii) enhance the FEM through meshing refinement around the pre-crack to improve accuracy for smaller cracks. The calibrated fatigue parameters will be implemented in 3D package-level simulations with explicit crack growth to evaluate damage at local stress concentrators (e.g., around solder interconnects) and validate against thermal-cycling experiments on simplified test vehicles. Finally, traction-separation laws derived from crack initiation tests will enable prediction of local crack nucleation in XFEM. Together, this approach enables material-specific package reliability assessment depending on lifetime models of materials.

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